Winter 2025

Advanced Drone Tracking Camera System

Jake Lamoureux | Laura Cao | Dominik Hoang | Kevin Pham

FUNDED BY LAB FUND ENGINEERING STUDENTS

Objective

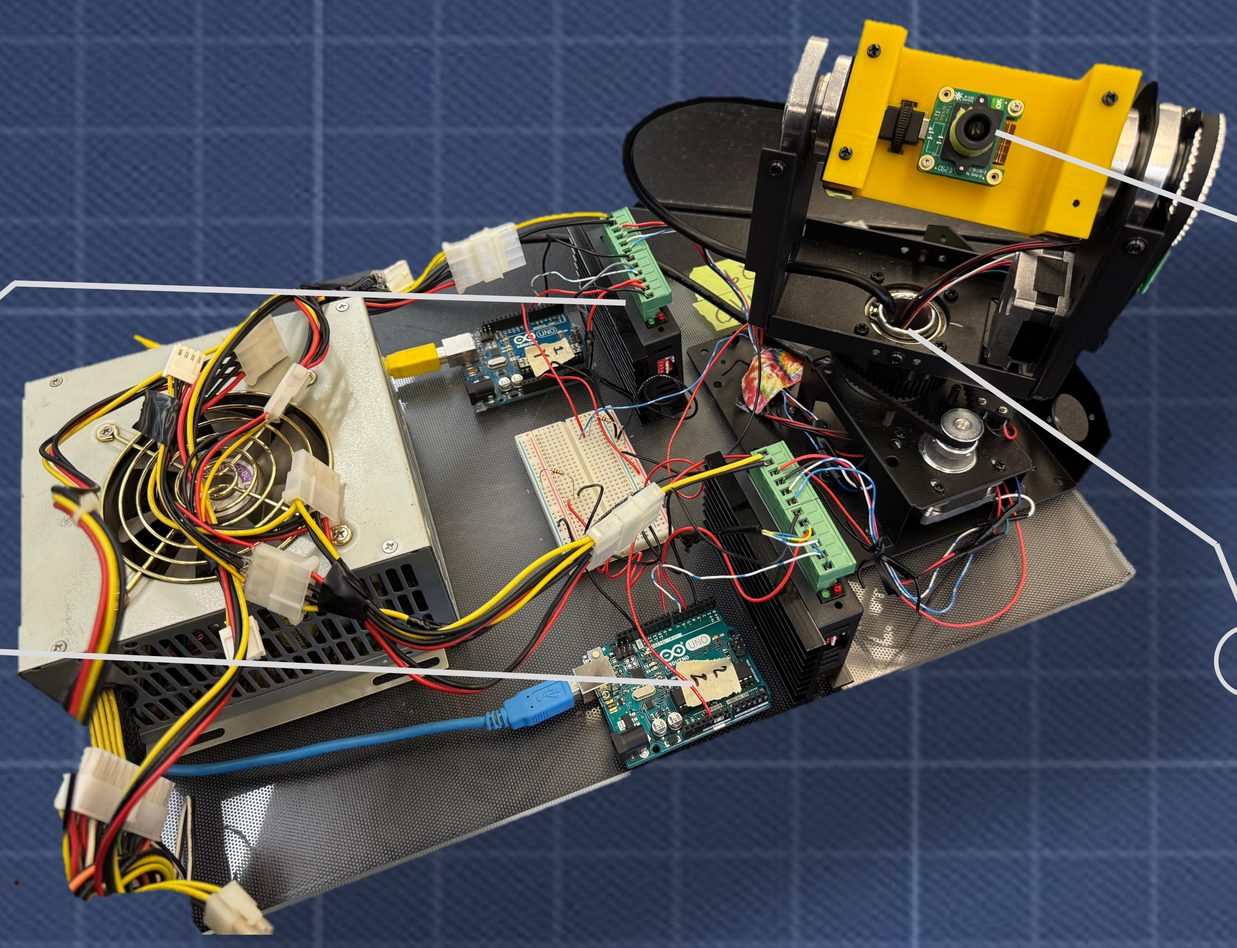
- Develop an affordable drone tracking system that provides real-time, accurate tracking while ensuring privacy for consumers and businesses.
- Integrate responsive servo motor control for smooth, precise tracking and reduced safety risks in dynamic environments.
- Optimize for power efficiency to lower operational costs and extend runtime, making the solution accessible to both commercial and household users.
- Design a scalable, adaptable system functional in various environments and affordable drone models.

Problem Statement

As drones become more widely used in homes and businesses, the need for effective tracking solutions that address privacy, real-time processing, and environmental adaptability becomes increasingly important. Many solutions are costly, inefficient, and fail to meet regulatory and safety standards. There is a clear need for an affordable, efficient, and privacy-conscious drone tracking system that integrates seamlessly into modern security frameworks while ensuring safety and compliance.

Background

- Drone use has grown rapidly, especially in consumer markets for home security and personal use.
- As drones become more widespread, privacy, safety, and effective tracking are growing concerns.
- While tracking systems have improved in accuracy, many still face challenges with real-time processing and adaptability.
- Existing solutions are often expensive and lack power efficiency for extended use.



Motor Controller

- TB6600 Stepper Motor Controller

Microcontroller

- Arduino Uno

120 FPS Camera

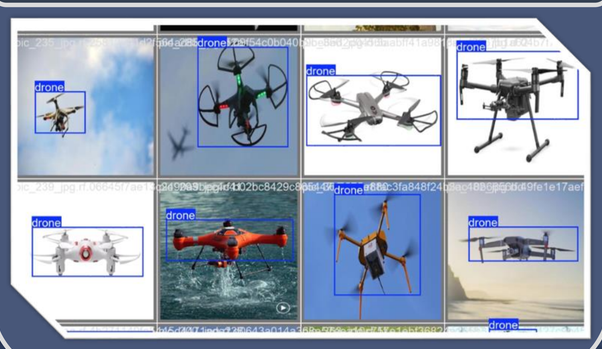
- Econ Systems See3CAM_24CUG

360° Camera Mount

- Two-stepper motor arm design for precise Y- and Z-axis movement.

Sample Output

Sample Dataset

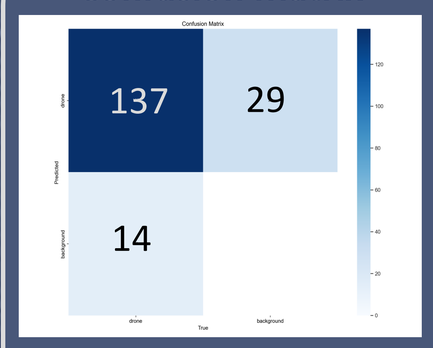


Design Solution

- The system integrates object detection technology for real-time tracking using visual processing.
- YOLOv8 was implemented to ensure high accuracy in dynamic environments.
- The design was tested under various lighting and movement conditions, balancing speed and accuracy.
- Servo motors control camera positioning, enabling smooth, responsive tracking with low latency.
- Feedback from drone operators and object detection experts refined the system's performance and reliability.
- The final design achieves accurate and fast tracking while maintaining regulatory compliance.

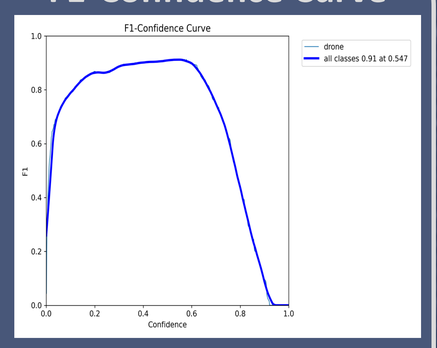
Results

Confusion Matrix



- 137 Correct drone detections (True Positives)
- 29 Background misclassified as drones (False Positives)
- 14 Missed drones (False Negatives)

F1-Confidence Curve



- F1 is a combination of Recall and Precision
- Max. Performance is at around 60% Confidence
- Higher = Strict (fewer detections & false alarms)
- Lower = Lenient (more detections & false positives)

UNIVERSITY of GUELPH

ENGINEERING

Faculty Advisor:
Hussein Abdullah, PhD, P.Eng.

COMPUTER ENGINEERING

ENGINEERING SYSTEMS & COMPUTING